Work Summary

This project will make it easier to detect uncommon behavior on smart-home devices and limit the chance of the device impacting other systems on the same network. Home and small office users don't have the tools and technical expertise to monitor IoT devices, particularly if a device that has been compromised seems to be functioning as usual.

To date the project has been able to document the network and its devices; it has captured traffic, covering around 24 hours before and after the segmentation events; and it has analysed the device destinations, protocols and traffic volumes. This information was used to establish behavioural baseline and detection rules. VLANs and firewall rules have been set up, and some connectivity tests were done, with the finding that any IoT-to-LAN attempts that were deemed to be prohibited were blocked. The team also looked into a Dropped Matter problem and fixed a connection-tracking timeout issue that was observed with this problem. Packet captures, analysis scripts, device profiles, logs and configuration records are all examples of supporting evidence.

The project offers a real-life workflow for analysing IoT behaviour with low cost tools. It correlates monitoring with network configuration and demonstrates how technical evidence can be utilized to explain suspicious

activity as well as normal device issues. The documented process can also serve as a reference for repeating similar assessments by students and small-network administrators.

The rest of the work is to complete the controlled large-volume upload test, measure false alarms on separate normal traffic, check device functions over longer time periods, and complete the Splunk dashboard, in preparation for the final portfolio. It should also be repeated by another practitioner to see if the instructions are understandable. The simulated lab could also be improved and be written into the report. Findings could be reinforced by longer captures and more consistent test conditions.